RIVER NAVIGATION ACADEMY

2-Day Student Lesson Plan

14	~9 hrs	5	I – IV
Sessions	On-water time	Core skill pillars	Class coverage

Skill pillars: Safety & PPE · Stroke Mechanics · Hydrology & River Reading · Navigation · Rescue

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Quick reference guide

INTERNATIONAL SCALE OF RIVER DIFFICULTY (ISRD)

Class I	Easy — fast water, small riffles, obvious channels. Self-rescue easy.
Class II	Novice — clear channels, some maneuvering, rare swimmer injury.
Class III	Intermediate — irregular waves, complex maneuvers, possible swamping.
Class IV	Advanced — powerful, predictable water requiring precise boat control.
Class V	Expert — violent, long, unavoidable drops. Rescue very difficult.
Class VI	Extreme / unrunnable — risk of life even with full preparation.

SIX CRITICAL RIVER HAZARDS

Strainer	Debris or branches that water passes through but a body cannot.
Undercut	Rock eroded below water level — entrapment risk on impact.
Sieve	Constriction allowing water but not boats or bodies through.
Hydraulic / Hole	Water flows over obstacle, curls back upstream; can recirculate.
Foot entrapment	Never stand in fast current — foot caught under rock.
Cold-water immersion	Raises effective difficulty of any class; impairs self-rescue.

RIVER FEATURES — READING WATER

Downstream V / Tongue	Deepest, fastest channel into rapid — primary navigation target.
Upstream V	Surface indicator of a submerged obstacle — avoid.
Eddy	Calm reverse-flow pool behind obstacle — rest and staging zone.
Pillow / Cushion wave	Foamy water on upstream face of rock — pressure wave.
Standing wave	Stationary wave; surfable but can indicate deeper hazard.
Eddy line	Boundary between main current and eddy — requires crossing angle.

W.O.R.M.S. SCOUTING FRAMEWORK

W	**Water**	Flow rate, depth, color — dark = deep/main channel
O	**Obstacles**	Rocks, strainers, undercuts — identify before launching
R	**Route**	Pick your line; mark entry, move, exit eddies
M	**Management**	Who is swimming? Who has throw bags? Safety positions?
S	**Safety**	Downstream hazards, bail-out eddies, rescue plan

Day 1 — Foundation

7:30 – 8:15 am Welcome, orientation & gear check SAFETY

LEARNING OBJECTIVES

- Establish group norms and expectations
- Review participant skill levels
- Fit PFDs, helmets, and river footwear
- Introduce rescue equipment: throw bags, knives

KEY CONCEPTS

◆ Type III / Type V PFD ◆ Helmet fit ◆ River knife placement ◆ Whistle signals (1 = all stop, 3 = emergency)

ACTIVITIES

- Introductions and skill-level survey
- PFD fitting station — each student verified by instructor
- Safety briefing: signals, swim posture, group protocol

✓ **Checkpoint:** Instructor sign-off: each student correctly fitted and able to demonstrate the 3-blast emergency whistle signal.

9:30 – 10:00 am *Break & gear to river*

8:15 – 9:30 am Whitewater class system & hazard identification THEORY

LEARNING OBJECTIVES

- Understand Class I–VI on the ISRD scale
- Recognize how flow, gradient, and obstacles combine
- Identify six critical river hazards
- Learn how water temperature elevates risk rating

KEY CONCEPTS

◆ ISRD Class I–VI ◆ Strainers ◆ Undercuts ◆ Sieves ◆ Hydraulics ◆ Foot entrapment ◆ Cold-water immersion

ACTIVITIES

- Classroom: ISRD scale breakdown with photo examples of each class
- Hazard ID exercise: rank a photo set of river features by risk level
- Local river map study — identify today's run class and hazard zones

✓ **Checkpoint:** Students correctly classify 8/10 rapid photos and identify 5 hazard types on the local map.

10:00 am – 12:00 pm Oar frame setup & stroke mechanics — flat water STROKE SKILLS

LEARNING OBJECTIVES

- Set up and adjust an oar frame correctly
- Execute catch, drive, finish, and recovery with proper sequencing
- Understand the 1:2 drive-to-recovery ratio
- Develop independent oar control (pull vs. push)

KEY CONCEPTS

◆ Oarlock over pivot point ◆ Catch angle ◆ Legs → back → arms sequence ◆ Feathering on recovery ◆ Forward vs. back stroke ◆ One-oar rudder ◆ Oar shipping safety

ACTIVITIES

- Oar frame rigging demo — students rig their own boats under supervision
- Flat-water drills: forward, back, and one-oar pivot on each side

- Paired feedback: one student rows, partner calls out sequencing errors
- Feathering drill — smooth oar rotation on each recovery pass

✓ **Checkpoint:** Instructors observe drive sequence: legs → back → arms, with clean catches and no blade washing out.

12:00 – 12:45 pm *Lunch on the river*

12:45 – 3:30 pm **Reading water — hydrology & river features** RIVER READING

LEARNING OBJECTIVES

- Identify eddies, tongues, pillows, holes, and waves from the bank
- Understand laminar vs. turbulent flow and the formation of each feature
- Apply the W.O.R.M.S. scouting framework
- Pick a navigational line before entering the boat

KEY CONCEPTS

◆ Downstream V / tongue ◆ Eddy formation & eddy line ◆ Pillow / cushion wave ◆ Hydraulic recirculation ◆ Standing waves ◆ W.O.R.M.S. framework

ACTIVITIES

- Bankside feature walk: instructor points out each feature on live Class II water
- Students mark features on a river sketch before running
- Row Class II reach — enter and exit three separate eddies per student
- Debrief each rapid: was the read correct? What was unexpected?

✓ **Checkpoint:** Each student successfully enters and exits three designated eddies using correct ferry angle, demonstrating tongue identification.

3:30 – 4:30 pm **Day 1 debrief & self-assessment** REVIEW

LEARNING OBJECTIVES

- Consolidate the day's learning
- Identify personal skill gaps
- Preview Day 2 curriculum

ACTIVITIES

- Students complete written self-assessment on stroke mechanics and feature identification
- Group debrief: one thing learned, one thing harder than expected
- Instructor preview of Day 2 — navigation, ferries, rescue

✓ **Checkpoint:** Completed self-assessment form collected; sets baseline for Day 2 instructor focus.

Day 2 — Application

8:00 – 8:45 am Morning review & swim safety / self-rescue SAFETY

LEARNING OBJECTIVES

- Review defensive swim position
- Understand foot-entrapment hazard
- Know when to abandon the boat
- Distinguish active vs. passive self-rescue

KEY CONCEPTS

◆ Defensive swim: feet up, toes up, downstream ◆ Aggressive swim: angling toward shore ◆ Foot entrapment — never stand in fast current ◆ Hydraulic escape: dive deep, swim to side

ACTIVITIES

- Bankside demo of defensive swim — students practice on calm shallows
- Foot entrapment scenario: instructor demonstrates, students identify hazard
- Scenario Q&A: what do you do first after flipping in a Class III hole?

✓ **Checkpoint:** Each student can verbally describe and physically demonstrate defensive swim posture and foot-entrapment awareness.

8:45 – 9:45 am Throw-bag rescue & shoreside rescue protocol SAFETY

LEARNING OBJECTIVES

- Accurately deploy a throw bag to a swimmer
- Perform a belay hold without tying off
- Set up downstream safety positions
- Understand rescuer-first principle

KEY CONCEPTS

◆ Throw bag — grab the rope, not the bag ◆ Pendulum effect to shore ◆ Belay position ◆ Never tie rope to yourself ◆ Two-rescuer assist (backup holder on PFD)

ACTIVITIES

- Throw-bag accuracy drill: hit a floating target from 30 ft, then 50 ft
- Live swimmer rescue pairs: one swims, one throws — rotate roles
- Set up a three-point downstream safety for a rapid — students assign positions

✓ **Checkpoint:** Students land throw bag within arm's reach of swimmer in 2 of 3 attempts; demonstrate correct belay position.

9:45 – 10:00 am *Break & gear to river*

10:00 am – 12:00 pm Ferry angles, pivots & eddy-hopping — Class II/III NAVIGATION

LEARNING OBJECTIVES

- Execute upstream and downstream ferries with controlled angle
- Perform stern pivot to use current for turning
- Chain eddy entries and exits across a rapid
- Maintain situational awareness while maneuvering

KEY CONCEPTS

◆ Upstream ferry: 45–90° angle, slower than current ◆ Downstream ferry: pulling through laterals ◆ Ferry angle vs. current speed tradeoff ◆ Stern pivot — engage feature to spin bow ◆ One-oar rudder for angle control

ACTIVITIES

- Ferry drill: cross river and land in target eddy — Class II advancing to III
- Stern pivot practice: approach feature broadside, pull downstream oar to spin
- Eddy-hop run: students must touch 5 specified eddies on a Class II/III reach
- Instructor review of ferry angle between laps

✓ **Checkpoint:** Students complete eddy-hop run with correct ferry technique in at least 4 of 5 target eddies.

12:00 – 12:45 pm *Lunch & scouting exercise at Class III rapid*

12:45 – 3:15 pm Navigating Class III — line selection & real-time adjustment APPLIED NAV

LEARNING OBJECTIVES

- Apply all skills in a continuous Class III run
- Select and execute a planned line
- Make real-time corrections when off-line
- Develop calm decision-making under current pressure

KEY CONCEPTS

◆ Pre-planned vs. reactive line selection ◆ Reading moving water from the boat ◆ Look-ahead: two moves ahead ◆ Consequence evaluation: swim probability per line

ACTIVITIES

- Guided Class III run — instructor leads first lap, students follow planned lines
- Solo laps: each student runs 2 independent laps with safety in place
- Between laps: verbal debrief — where did you deviate and why?
- Optional: instructor photos key sections for stroke and line review

✓ **Checkpoint:** Each student completes 2 solo laps with no unintended swims; instructor rates line selection quality on a 1–5 scale.

3:15 – 4:30 pm Final debrief, skills assessment & next steps REVIEW

LEARNING OBJECTIVES

- Summarize two-day skill progression
- Issue individual written feedback
- Set a personal development plan
- Cover resources for continued learning

ACTIVITIES

- Written instructor assessment: stroke mechanics, river reading, safety, navigation
- Individual feedback session: strengths and one priority to develop next
- Recommended resources: American Whitewater, ACA courses, local club programs
- Program certificate and group photo at take-out

✓ **Checkpoint:** Completed assessment form returned to each student; instructor retains copy for program records.